

Set 23: Moles, particles and mass

Notes

Atoms, molecules and ions are sub-microscopic with incredibly small masses. Yet we measure their mass and handle atoms in the laboratory all the time. So how do we do it? Small particles are dealt with in large numbers. We don't measure sugar and sand out as single grains. We use sensible units like teaspoons of sugar and truckloads of sand. Atoms are dealt with in a similar way where the sensible unit is called the mole. A mole of anything contains 6.022×10^{23} particles.

The number of atoms in 12.00 g of carbon-12 is 6.022×10^{23} . This very, very large number, 6.022×10^{23} is called Avogadro's number. Avogadro's number of particles is called a mole. Therefore one mole of any substance will contain 6.022×10^{23} atoms, molecules or formula units. The mole is a unit of measure used in chemistry.

For example:

- A mole of iron (Fe) consists of 6.022×10^{23} atoms of iron.
- A mole of carbon dioxide molecules (CO_2) consists of 6.022×10^{23} molecules of carbon dioxide.
- A mole of copper(II) sulfate (CuSO_4) consists of 6.022×10^{23} formula units of copper(II) sulfate.

Moles & mass

The relationship between the number of moles (n) and mass of a substance (m), in grams, is:

$$n = \frac{m}{M}$$

where M = molar mass of a substance in g mol^{-1} .

Examples

1. How many moles of calcium atoms are there in 30.0 g of calcium?

$$\begin{aligned} n &= \frac{m}{M} \\ &= \frac{30.0}{40.08} & M(\text{Ca}) &= 40.08 \text{ g mol}^{-1} \\ n &= 7.49 \times 10^{-1} \text{ mol of Ca atoms} \end{aligned}$$

2. How many moles of benzene (C_6H_6) molecules have a mass of 390.0 g?

$$\begin{aligned} n &= \frac{m}{M} \\ &= \frac{390.0}{78.107} & M(\text{C}_6\text{H}_6) &= 78.107 \text{ g mol}^{-1} \\ n &= 4.99 \text{ mol of C}_6\text{H}_6 \text{ molecules} \end{aligned}$$

3. Calculate the mass of 5.0 moles of barium sulfate (BaSO_4) formula units.

$$\begin{aligned} n &= \frac{m}{M} \\ m &= nM \\ &= (5.0)(233.36) & M(\text{BaSO}_4) &= 233.36 \text{ g mol}^{-1} \\ m &= 1.2 \times 10^3 \text{ g} \end{aligned}$$

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Set 23: Exercises

- Calculate the number of moles contained in each of the following:
 - 72.0 g of magnesium (Mg)
 - 4.00×10^2 g of calcium carbonate (CaCO_3)
 - 104 g of ethyne (C_2H_2)
- Calculate the mass of each of the following:
 - 4.75 moles of lithium atoms
 - 0.25 moles of sodium hydroxide formula units
 - 9.00 moles of carbon monoxide molecules
- Calculate the number of moles of molecules or formula units contained in each of the following:
 - 28.0 g of nitrogen
 - 232 g of butane (C_4H_{10})
 - 3.90 g of sodium peroxide (Na_2O_2)
- Calculate the number of moles of hydrogen peroxide (H_2O_2) molecules in 119 g of hydrogen peroxide.
- Calculate the number of moles of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) molecules in a can of cool drink that contains 39.8 g of sucrose.
- Calculate the number of atoms in the following:
 - 1.00 mole of gold (Au)
 - 0.25 moles of sodium (Na)
 - 2.33 moles of chlorine gas (Cl_2)
 - 0.0894 moles of phosphorus (P_4)
- Calculate the total number of ions in each of the following:
 - 1.11 moles of sodium chloride (NaCl)
 - 2.98 moles of aluminium oxide (Al_2O_3)
 - 1.11×10^{-5} moles of magnesium fluoride (MgF_2)
 - 0.222 moles of calcium sulfate (CaSO_4)
- Calculate the total number of atoms in each of the following:
 - 0.189 moles of barium phosphate ($\text{Ba}_3(\text{PO}_4)_2$)
 - 25.0 g of bromine (Br_2)
 - 0.0678 g of nitric acid (HNO_3)
 - 12.5 g of acetic acid (CH_3COOH)